ISO-1 continuation (2026-10-16): certified labels, the 8th-power mass, and the residual zero classes

Research evidence report. Study: `research/iso1_weak_classes_20261007`. Markdown version with links: `residual_model_20261016/REPORT.md`. Branch `aburan28/quirky-feynman-prppv2`, PR aburan28/crypto#1954. Host: 4-core Linux container, shared; wall times are not isolated measurements.

Summary

1. **Certified census labels through p = 97.** A new counter returns a group order only when it is the unique Hasse-interval candidate compatible with the tested points on the curve and its quadratic twist. Reruns at p = 7, 13, 37, 41, 43, 47, 53, 59 are byte-identical to the retained CSVs, so their zero labels are exact. p = 61, 67, 71, 73, 79, 83, 89 and 97 are new.
2. **Exact 8th-power mass (conjecture).** The number $N_8(t)$ of 8th-power Legendre parameters in a class is a closed 2-adic factor times an odd-conductor Hurwitz number. It matches exhaustive censuses at p = 7, 11, 13, 17 and the registered holdouts p = 19 and p = 23 with 0 mismatches.
3. **Random thinning.** Beyond that structure, the norm-one torus samples each class at random. The parameter-free model predicts the residual zero count at the pristine holdouts p = 53, 59 and at the forward predictions p = 61, 67, 71, 73, 79, 83, 89, 97, with binomial dispersion 1.00 ± 0.01.

Beyond the proved obstruction $v_2(f_\pi) \geq 2$, whether a class is weak is a draw with probability $1 - \left(1 - \frac{8}{p^2-1}\right)^{N_8 \frac{t}{12}}$. The evidence does not support any compact deterministic criterion beyond that.

1. Certified point counts

$Q = p^6$, so the Hasse interval is exactly $[Q + 1 - 2p^3, Q + 1 + 2p^3]$. Every curve has full rational 2-torsion, so $4 \mid \#E$, and the twist has order $2Q + 2 - \#E$. A full baby-step/giant-step scan of $4P$ lists every candidate. Further points on the curve and the twist filter the list, and the order is returned only when one candidate remains. Source: `census_runtime/src/certified.rs`. Unit test: 12 random p = 7 curves against brute-force counts.

p	point counts	extra points	wall s	result
7	213	5	0.01	identical to exact square-table census
13	2,423	8	0.2	identical to retained CSV
37	156,523	16	32	identical
41	235,901	20	56	identical
43	285,363	27	110	identical
47	407,193	35	118	identical
53	658,243	27	347	identical
59	1,010,651	23	941	identical to retained p59 run
61	1,154,751	41	1,265	new: 0 / 111,630 depth-1 weak; 546 / 111,630 high-depth zero
67	1,680,383	30	1,448	new: 0 / 148,138 depth-1 weak; 640 / 148,136 high-depth zero
71	2,118,901	34	resumed	new: 0 / 176,436 depth-1 weak; 634 / 176,434 high-depth zero
73	2,367,853	27	resumed	new: 0 / 191,844 depth-1 weak; 622 / 191,844 high-depth zero
79	3,247,401	35	resumed	new: 0 / 243,400 depth-1 weak; 746 / 243,398 high-depth zero
83	3,956,583	33	resumed	new: 0 / 282,450 depth-1 weak; 798 / 282,448 high-depth zero
89	5,230,501	53	resumed	new: 0 / 348,524 depth-1 weak; 808 / 348,524 high-depth zero
97	7,379,793	39	resumed	new: 0 / 451,632 depth-1 weak; 934 / 451,632 high-depth zero

Receipts: `certified_20261016/` (comparison receipt, stderr receipts, CSV hashes, queue script and status).

2. Exhaustive λ census and the 8th-power mass

`iso1_lambda_census` enumerates every λ in F_Q or in a 2-power subgroup. For each it records the certified trace of $y^2 = x(x-1)(x-\lambda)$, whether λ is a 2^k-th power, and whether it lies in the norm-one torus T (the weak curves).

- At p = 11 it agrees with the orbit census on all 666 rows. In each class, membership in T thins $\Omega = \left(F_Q^*\right)^{2^e}$ at random: dispersion 1.012, with 31 zero classes against 32.4 expected.

- 32nd powers at p = 17 are not 2-adically local; 8th powers are, at every prime tested. This matches the fact, quoted here and not reproved, that the cover $\lambda = \mu^N$ is a congruence modular curve only for $N \mid 8$.

With $\Delta = t^2 - 4Q = f_\pi^2 D_K$, $k = v_2(f_\pi) \geq 2$, $s = \left(\frac{D_K}{2}\right)$ and H_{odd} the Hurwitz sum with the 2-part of the conductor removed:

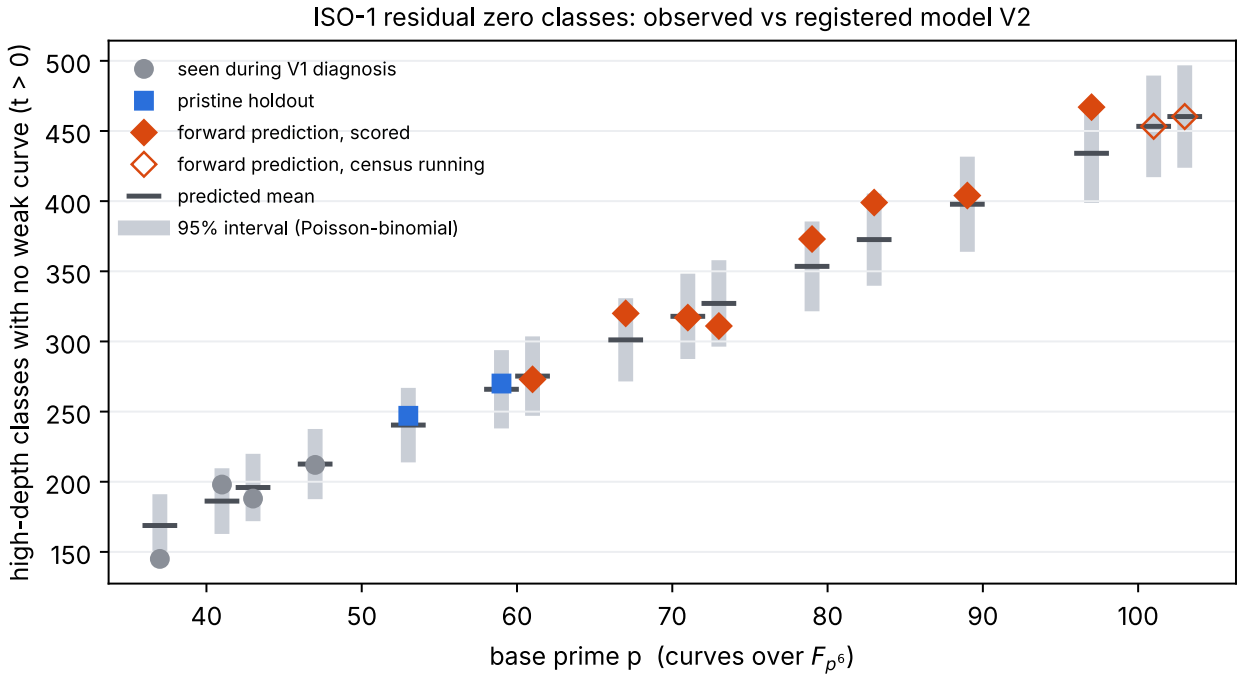
$$N_8(t) = H_{\text{odd}(t)} \cdot \begin{cases} (2-s)2^{k-2} & \text{if } t \not\equiv \pm(Q+1) \pmod{32} \\ 5(2-s)2^{k-2} - 6(1-s) & \text{if } t \equiv \pm(Q+1) \pmod{32} \end{cases}$$

test	primes	rows	mismatches
derivation (exhaustive 8th-power census)	7, 11, 13, 17	4,078	0
registered exactness holdout	19	3,249	0
registered exactness holdout	23	5,819	0

The formula is read off the tables and is not proved.

3. Residual zero classes: registered model V2

V1, which used class size only, was registered and failed on its training primes (dispersion about 5). V2 uses only small-prime information: weak orbits follow Binomial$\left(\frac{N_8}{12}, \frac{8}{p^2-1}\right)$, with nothing fitted. It is evaluated natively by `iso1_residual_model`.



Certified censuses (certified_20261016); predictions from class data only. Source: residual_model_20261016/model_v2_results.txt, predictions_v2.txt

Figure 1: Observed high-depth zero classes ($t > 0$) against registered V2 predictions with 95% Poisson-binomial intervals. Filled markers: certified censuses (grey: seen during V1 diagnosis; blue: pristine holdouts; orange: forward predictions, scored). Hollow: forward predictions whose census is running. Data: `figures/zero_rows.csv`; sources `model_v2_results.txt`, `predictions_v2.txt`.

p	role	rows	zeros	predicted [95%]	disp.	AUC
37	seen	12,321	145	168.8 [146.5, 191.1]	0.981	0.959
41	seen	16,810	198	186.2 [162.8, 209.6]	0.998	0.969
43	seen	19,414	188	195.9 [171.9, 219.9]	1.003	0.972
47	seen	25,403	212	212.6 [187.6, 237.6]	0.997	0.978
53	pristine holdout	36,517	247	240.4 [213.8, 266.9]	1.005	0.982
59	pristine holdout	50,474	270	265.9 [238.0, 293.8]	1.001	0.986

p	role	rows	zeros	predicted [95%]	disp.	AUC
61	forward	55,815	273	275.3 [247.0, 303.6]	0.992	0.984
67	forward	74,068	320	301.1 [271.5, 330.8]	0.996	0.987
71	forward, e = 4	88,217	317	317.9 [287.5, 348.3]	1.000	0.989
73	forward, e = 4	95,922	311	327.1 [296.3, 357.9]	1.011	0.989
79	forward, e = 5	121,699	373	353.5 [321.5, 385.5]	0.998	0.993
83	forward, e = 3	141,224	399	372.6 [339.7, 405.4]	1.001	0.993
89	forward, e = 4	174,262	404	397.8 [363.9, 431.7]	1.000	0.995
97	forward, e = 6	225,816	467	434.1 [398.7, 469.6]	1.001	0.996

p = 37 lies 1.5 below its interval; every other prime is inside. p = 97 (467 against 434.1) sits just inside the upper edge of its interval. Pooled over the ten scored primes 53–97, observed minus predicted is +95.3 with standard deviation 49.8 (z = 1.91, about 1.9% of the expected total). The excess is in the bulk of the predicted-probability range, not the tails, and it is not isolated by e = $v_2(p^2-1)$: e = 3 gives z = 1.62, e = 4 gives −0.39, e = 5 gives 1.19, e = 6 gives 1.82, and e ≥ 4 gives 1.12. That refutes the idea that it comes from the 8th-power model being inexact at larger e. The excess is borderline and was noticed in scored data, so it is registered as a follow-up test on fresh primes 101, 103, 107, 109 and 113 (PROTOCOL_V2.md, commit ff9e0a3f8, made before p = 101 completed): V2 is rejected as a parameter-free description if their pooled one-sided z exceeds 2.0. h_K alone ranks zero rows with AUC 0.71–0.78.

4. Scope and limits

- **Proved** (`THEOREM.pdf`): no ordinary class with $v_2(f_\pi) = 1$ is weak, for every odd p.
- **Exact** for $p \leq 97$: the certified class labels.
- **Conjecture**, exact on six primes: the N_8 formula.
- **Measured**: the binomial thinning model, at two pristine holdouts and eight forward predictions. It describes the data statistically and decides no individual class. It does not show that no exotic arithmetic criterion exists.
- Class numbers come from PARI/GP 2.15.4 `quadclassunit`, which is GRH-conditional for large $|D_K|$. This affects the predictions, not the census labels.
- Pending: the censuses for p = 101–199 and the registered follow-up test. p = 71–97 ran across several container sessions with checkpointing, so it has no single wall time. p = 199 alone needs about 4 days on 4 threads.